

# Can Blood Vessel Networks in Ultrafast Breast MRI Predict Breast Cancer Treatment Response?

Spencer Venancio<sup>1</sup> Sarit Bose<sup>2</sup> Aakrithi Ram<sup>3</sup> Advisor: Anna Woodard, PhD<sup>4</sup>

<sup>1</sup>University of Wisconsin–Madison

<sup>2</sup>Neuqua Valley High School

<sup>3</sup>Bergen County Academies

<sup>4</sup>Data Science Institute, University of Chicago

## Why Might Blood Vessels Predict Treatment Response?

Neoadjuvant treatment (NAT) is given before breast cancer surgery. **Pathologic complete response (pCR)**—no residual invasive cancer at surgery—is associated with better outcomes [1]. Earlier prediction could support treatment tailoring: less intensive therapy for patients likely to respond well and alternative therapy for those unlikely to achieve pCR.

Tumors recruit abnormal vessels that can be dense, tortuous, leaky, and disorganized [3]. Their structure and contrast-enhancement dynamics may reflect tumor biology and affect drug delivery. We test whether these vessel-network features improve pCR prediction beyond simple summaries of the same MRI measurements.

**Research Question:** does the structure of breast blood vessel networks and the way contrast changes through them improve response prediction beyond simple summaries of the same MRI measurements?

## UChicago Cohort: 179 Labeled, 1,226 Unlabeled Exams

UChicago contributed **181 labeled exams from 143 patients; 179 exams from 141 patients** are analyzed here; **48 of the 141 patients (34%)** achieved pCR. Two exams lacked the required centerline, support mask, and aligned MRI.

Acquisition timing is heterogeneous: exams contain 13–24 frames spanning 96–480 s. Longitudinal exams are included. This analysis measures pCR discrimination across longitudinal exams; baseline-only early prediction remains to be tested. Cross-validation is grouped by patient, so one patient's exams never cross training/evaluation splits.

Pretraining draws on a separate, non-overlapping pool: **1,226 unlabeled exams from 225 patients**, with no patient or exam shared with the labeled cohort.

## Three Graph Representations Test Whether Topology Matters

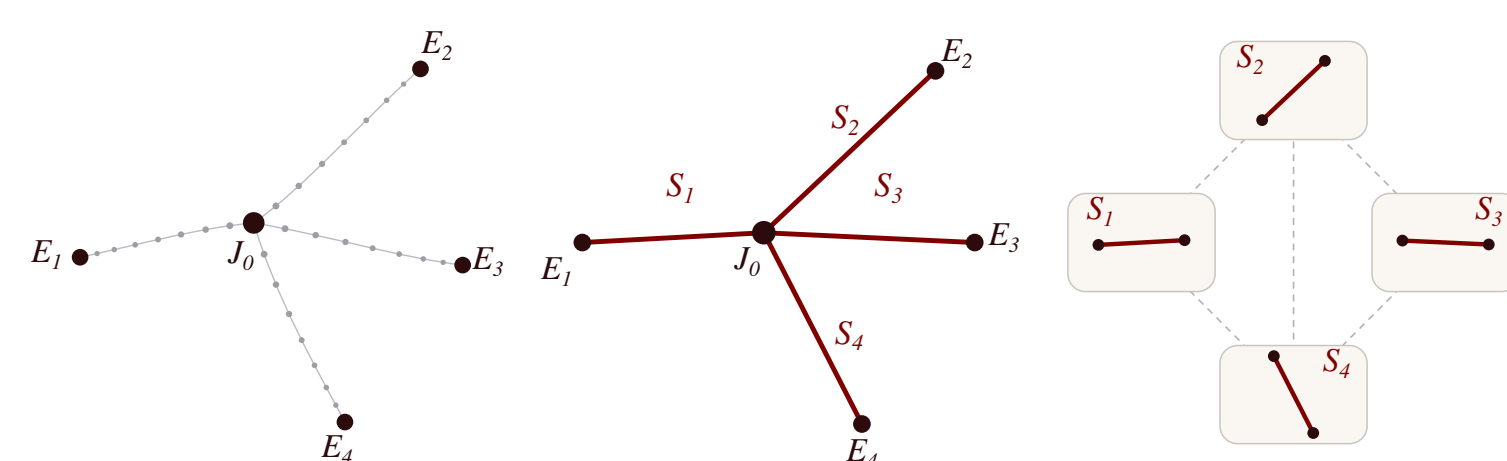


Figure 1. The same vessel centerline represented three ways.

We represent the vascular network of each patient's breasts as a graph:

- **Voxel graph:** each vessel-centerline voxel is a node; neighboring centerline voxels are connected.
- **Junction graph:** branch points are nodes ( $v \in V_{\text{vox}}$ ,  $\deg(v) > 2$ ); vessel segments connect them.
- **Segment graph:** vessel segments are nodes; segments sharing an endpoint are connected.

## Vessel Centerlines Become Graphs for pCR Prediction

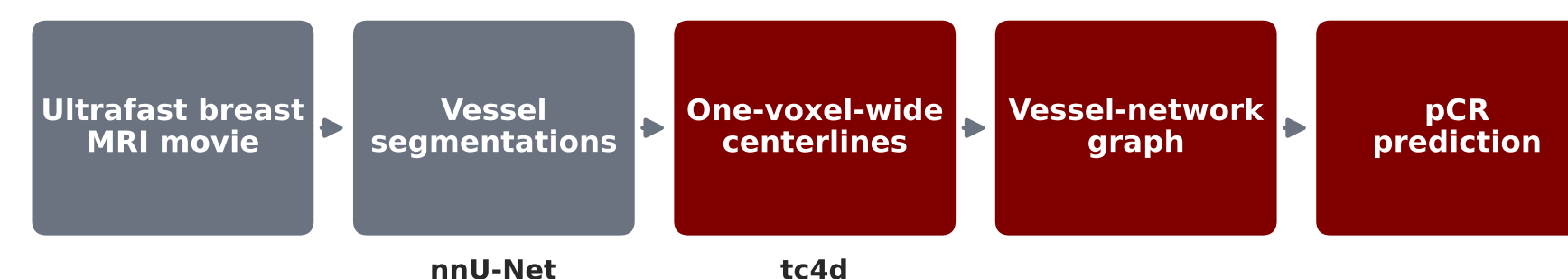


Figure 2. Ultrafast breast MRI movie to vessel-network prediction.

A published deep-learning method (nnU-Net) identifies vessel voxels [2]. Our **tc4d** algorithm (temporal-consensus 4D, using consistency across DCE-MRI time points to extract an exam-level vessel skeleton) converts that mask into a one-voxel-wide centerline while preserving vascular connectivity. The centerline then becomes one of the three graph representations.

Nodes carry geometry and measured enhancement curves. A **graph neural network (GNN)** combines neighboring-node information, pools the network into one exam representation, and predicts pCR.

## Forecasting Pretraining: A Controlled 2x2 Comparison

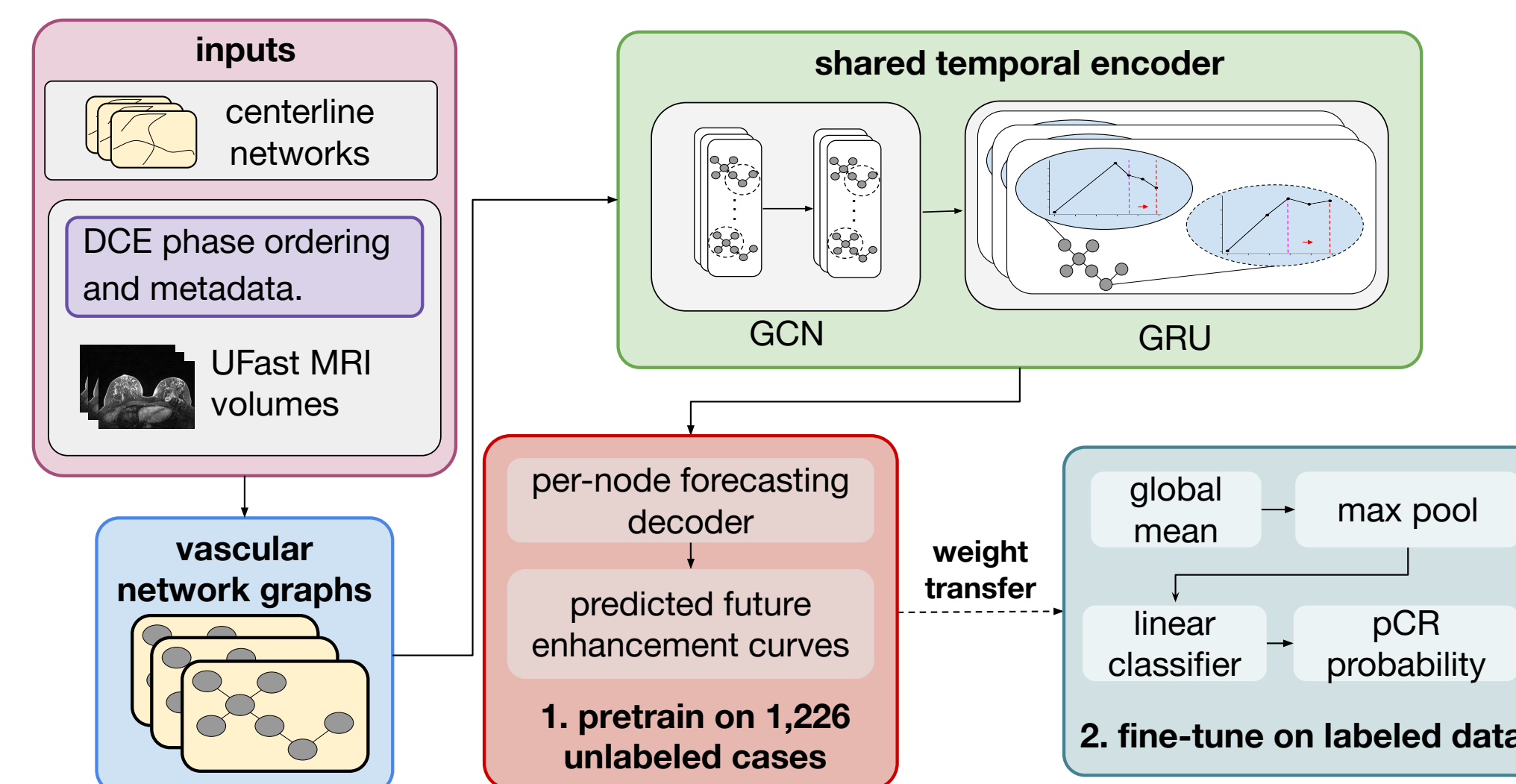


Figure 3. Forecasting weights transfer into the pCR classifier; a randomly initialized twin is the control arm.

The encoder receives vessel curves, connections, and elapsed times and predicts **future enhancement at connected vessel nodes** [4]. The downstream comparison holds inputs, encoder, training, patient-grouped folds, and evaluation fixed: forecasting versus random initialization, with and without graph message passing.

Centerline enhancement features predict pCR modestly, but hand-engineered GNNs do not beat matched tabular summaries—motivating this pretraining comparison as an alternative route to a better node representation.

## Forecast Pretraining Improves the GNN

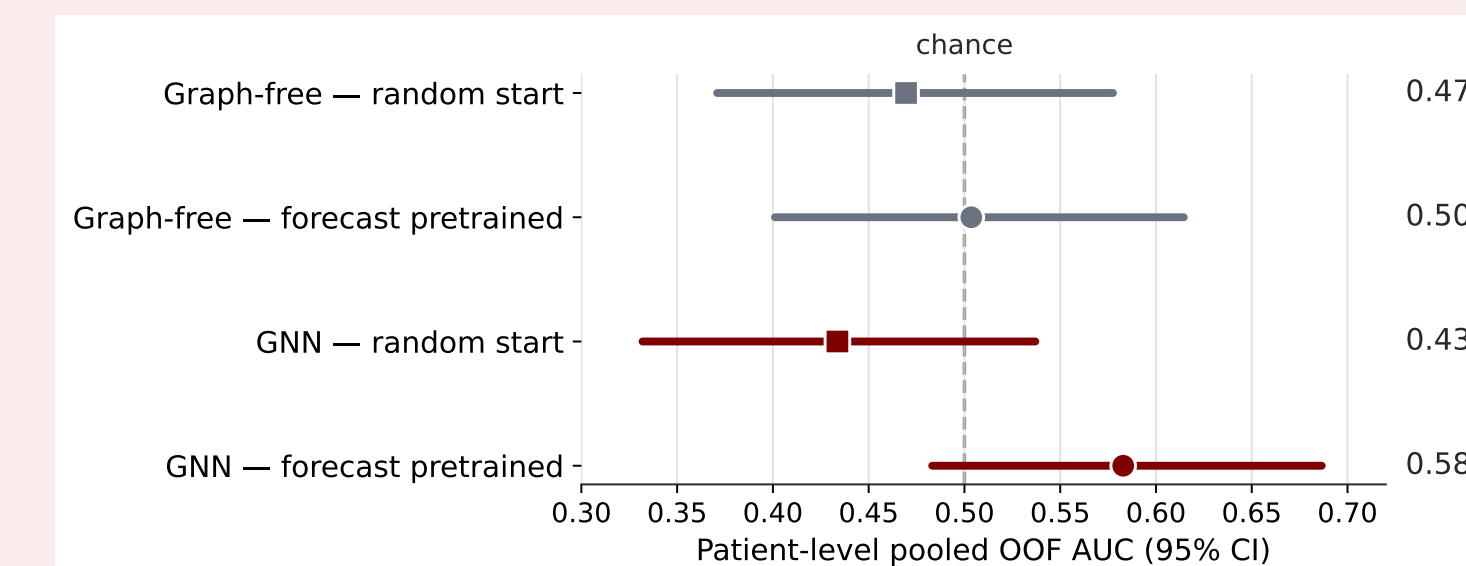


Figure 4. Patient-level pooled out-of-fold AUC (each patient's exam probabilities averaged to one prediction) with patient-bootstrap 95% CIs on shared patient-grouped folds.

The GNN uses voxel graphs (one node per centerline voxel), matching the existing pipeline rather than swept against segment/junction representations.

Paired pretraining  $\Delta\text{AUC}$ : **+0.149 [+0.046, +0.255]** for the GNN, versus +0.034 [−0.049, +0.120] for graph-free (95% CI crosses zero). The pretrained GNN exceeded pretrained graph-free by  $\Delta\text{AUC}$  +0.079 [+0.010, +0.155].

## Prediction Is Sensitive to Acquisition Timing

Acquisition timing is associated with outcome in this cohort: a model using frame count, duration, and cadence without imaging features reaches **0.587 AUC**—essentially the pretrained GNN's 0.583. Their predictions correlate (Pearson  $r = 0.50$ ).

This raises concern about protocol dependence, but cannot separate technical from biological association. Adjustment can remove genuine patient differences that happen to track acquisition as well as technical variation.

Forecast pretraining improved the vessel GNN, which outperformed the matched graph-free model. Acquisition timing was also associated with outcome, motivating follow-up analyses to distinguish protocol-related and vascular contributions to prediction.

## Acknowledgments + References

We thank Zhen Ren, Milica Medved, and Greg Karczmar for feedback and MRI-physics expertise; Fred Howard for the development cohort; and Bella Summe, Julia Luo, José Cardona Arias, and Rebecca Wu for earlier vessel-pipeline development. HIRO (RRID:SCR\_018372) coordinated imaging (Ludwig Fund; NIH/NCI P30 CA014599). Randi HPC/CRI compute was supported by BSD and ITM/CTSA NIH UL1TR002389.

- [1] Cortazar et al. *Lancet*. 2014;384:164–172.
- [2] Isensee et al. *Nature Methods*. 2021;18:203–211.
- [3] Nagy et al. *Br J Cancer*. 2009;100:865–869.
- [4] Tang et al. *Proc AAAI*. 2022;36:12289–12297.